# Multiscale Spatial-Frequency Domain Dynamic Pansharpening of Remote Sensing Images Integrated With Wavelet Transform

Yingjie Li, *Graduate Student Member, IEEE*, Weiqi Jin, Su Qiu, and Yuqing He

***Abstract*— Pansharpening produces high-resolution multispectral (HRMS) images by enhancing the high-frequency details in multispectral (MS) images, with deep learning (DL) methods emerging as a new paradigm. However, most existing DL methods primarily focus on pansharpening in the spatial domain using convolutional layers with local kernels, which is not conducive to accurate reconstruction of global high frequencies, leading to spatial deformation and spectral distortion. To address this issue, we propose a bidirectional progressive spatial-frequency dynamic fusion network that constructs a multiscale structure using wavelet transform. This network utilizes the wavelet transform's ability to analyze frequency information in local neighborhoods to inject high frequencies from panchromatic (PAN) images into MS images. Additionally, it employs a wavelet-based adaptive fusion module to analyze spatial–spectral correlations and adaptively learn the injection gains for high-frequency details, thereby enhancing spectral fidelity. To further adapt to various scenarios, a dual-domain dynamic filtering (D3F) block is proposed. It improves model generalization through data-specific dynamic filtering in both frequency and spatial domains, capturing global and local dependencies effectively. Quantitative metrics and visualization results on reduced and full-resolution images demonstrate that our method outperforms other advanced methods with low-computational cost, producing high-fidelity HRMS images.**

***Index Terms*— Deep learning (DL), dynamic filtering, frequency domain, pansharpening, wavelet transform.**

## I. Introduction

REMOTE sensing images with high spatial and spectral resolution have broad application prospects, including land use mapping [1], [2], urban planning [3], environmental protection [4], [5], safety monitoring [6], and 3-D reconstruction [7]. However, acquiring such images remains a significant challenge. When the incident energy remains constant and the number of spectral bands increases, the energy allocated to each band decreases. Given that the satellite orbit altitude is preset, the instantaneous field of view needs to be increased to capture sufficient energy [8]. This requirement directly influences the size of the ground-resolution cell, thereby resulting in reduced spatial resolution. Pansharpening technique thereby has been developed to produce high-resolution multispectral (HRMS) images by fusing multispectral (MS) and panchromatic (PAN) images that respectively possess high spectral and spatial resolution.

Traditional research on pansharpening primarily treated it as either the injection of spatial information from PAN image into MS image—such as in component substitution (CS) methods [9] and multiresolution analysis (MRA) methods [10]—or as a model-optimization problem, like variational optimization (VO)-based methods [11]. However, these traditional methods inevitably result in spatial and spectral distortions, due to their reliance on prior knowledge assumptions and handcrafted features [12], [13]. In contrast, deep-learning (DL) techniques, especially convolution neural networks (CNNs), have demonstrated exceptional performance in feature learning and have become a new paradigm for pansharpening. Masi et al. [14] proposed a CNN-based pansharpening method by tuning the super-resolution network SRCNN [15]. Following this work, a multitude of DL-based methods have emerged. Yang et al. [16] proposed PanNet with ResNet as the backbone to enforce structural consistency in the high-pass domain. Liang et al. [17] extracted MS features and the residuals between MS and PAN images by two branches with multireceptive-field attention. Wu et al. [18] proposed cross-scale parallel branches to capture MS features and fused them with prefused features via dynamic weights. However, most DL-based methods focus on spatial domain pansharpening using convolutional layers with local kernels, which are limited in establishing global representations. This limitation can lead to the neglect of the overall structure during the reconstruction of high-frequency details, thereby resulting in spatial detail blurriness and local distortions in the generated HRMS image.

In contrast, frequency domain analysis excels at extracting and processing high-frequency information in a holistic perspective of the image, which is conducive to improving the spatial and spectral fidelity of the HRMS images produced. From a frequency domain perspective, pansharpening fundamentally involves filling the high-frequency gaps in MS images under the guidance of the PAN image, a process that can be facilitated by wavelet transform. To illustrate this, we analyze pansharpening in the frequency domain in Fig. 1, where the magnitude spectrums of PAN, HRMS, and upsampled MS (denoted as LMS) images are depicted. The magnitude spectrums reveal that the energy of the LMS image is concentrated mainly in low frequencies, while PAN

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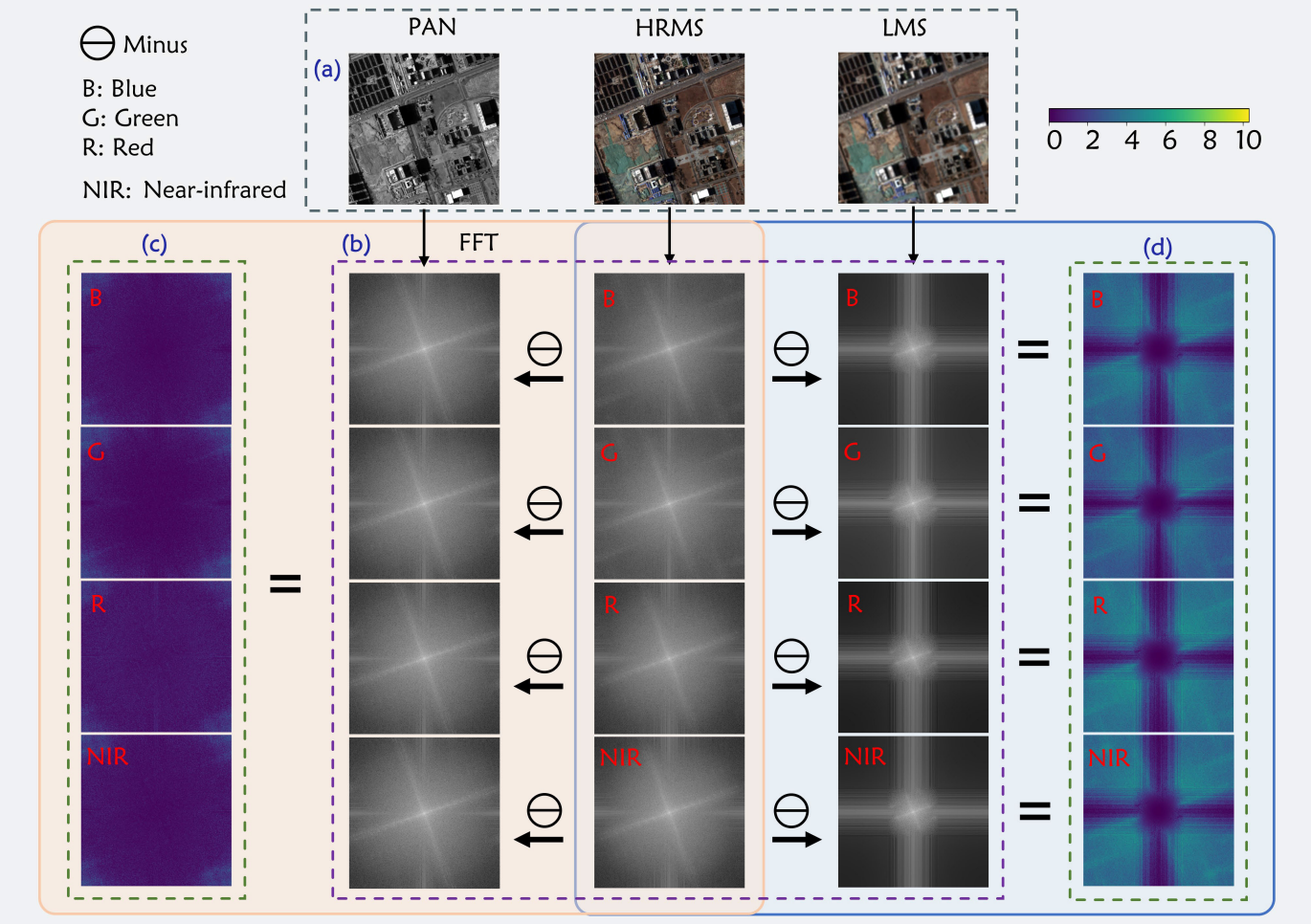



Fig. 1. Frequency domain analysis of pansharpening. (a) LMS, PAN, and HRMS images. (b) Magnitude spectrum of HRMS, PAN, and LMS. (c) Residual maps of magnitude spectrum between PAN and each band of HRMS. (d) Magnitude residual maps between each band of HRMS and LMS.

and HRMS images also contain a wealth of information at high frequencies. To better depict the frequency domain relationship between HRMS, PAN, and LMS, we also present the magnitude residual maps between PAN and HRMS, as well as LMS and HRMS. Fig. 1(c) and (d) depicts that LMS lacks information at high frequencies compared to HRMS, while the high-frequency magnitude spectrums of PAN and HRMS images exhibit similar energy distributions. Therefore, HRMS images can be effectively generated by filling the high-frequency information in the PAN into the MS image. Wavelet transform, as a primary tool for image frequency domain analysis, decomposes an image into different frequency sub-bands at multiple scales, capturing both spatial and frequency information [19], [20]. This makes it excel in precise frequency analysis and can facilitate the injection of decomposed PAN high-frequency information into MS images. However, as depicted in Fig. 1(c), the residual maps of HRMS and PAN images differ in different bands, thus, the high-frequency information cannot be directly transferred. Instead, a guided strategy should be adopted to appropriately integrate these frequencies. This strategy needs to account for the implicit relationships between different bands of MS images and those between PAN and MS images. Parsing these relationships can be challenging, but can be effectively achieved by DL models. However, there is currently limited research on integrating the advantages of wavelet transform with DL models for pansharpening.

In this work, we enhance pansharpening by applying the multiscale decomposition capability of wavelet transforms, which is compatible with the multiscale structure of DL models. We proposed a bidirectional dynamic filtering network (BiDFNet) that includes both reverse (decomposition) and forward (reconstruction) processes, enabling the progressive reconstruction of HRMS images at multiple scale levels. Initially, in the reverse network, the wavelet transform decomposes PAN features into approximation and high-frequency detail components across multiple scales until they match the scale of the MS features. Specifically, the scaling functions of the wavelets replace traditional pooling layers in DL models, extracting multiscale features by computing the low-frequency approximation components of an image. Since the scale functions maintain the image frequency characteristics during the pooling process [21], [22], [23], the multiscale features preserve more long-distance relevant information. Furthermore, the basic functions of the wavelet extract various levels of high-frequency components, allowing for the independent analysis of specific components. These components, embed both local details and global representations, facilitating the

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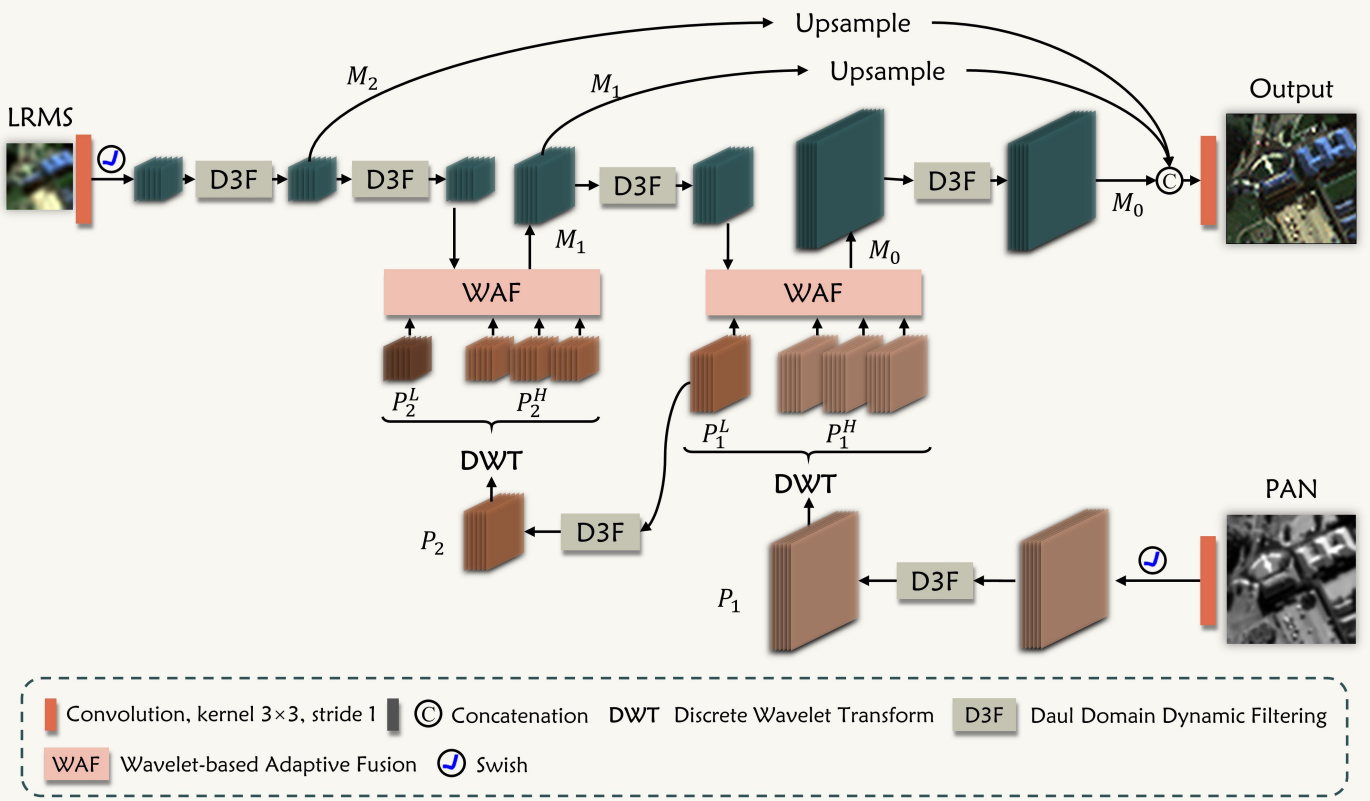



Fig. 2. Overall architecture of BiDFNet.

progressive injection of high-frequency details. Subsequently, in the forward network, these components are fused with the MS features through inverse wavelet transform, guided by band-specific injection gains learned from spatial–spectral dependencies. By taking the original MS images as inputs, this method mitigates the spectral distortion caused by directly upsampling MS images before model input, achieving a stepwise enhancement in the resolution of MS features. Additionally, the differences in feature scale and spectral responses across different images present challenges to model generalization, requiring data-specific processes to increase the method's adaptability. To address this, we introduce dynamic filtering in both the spatial and frequency domains. This enables the model to adaptively learn from different scenarios and thereby generate images with high fidelity in both spatial and spectral aspects.

The main contributions of this article can be summarized as follows.

1) We propose a novel bidirectional progressive pansharpening scheme constructed which integrates wavelet transform with DL model, to extract high-frequency details from PAN features at multiple scale levels and subsequently inject these details into MS features, adeptly avoiding the introduction of spectral noise commonly associated with the direct upsampling of MS images.
2) We present a wavelet-based adaptive fusion (WAF) module, which reconstructs HRMS features through inverse wavelet transform and adaptively learns band-specific injection gains in each band based on spatial–spectral dependencies across PAN and MS features.
3) We design a dual-domain dynamic filtering (D3F) block as basic blocks in the backbone, which combines frequency domain filtering and spatial domain filtering to establish global and local representations, and to improve the generalization of the model by adaptively adjusting filter weights according to the scenarios.

## II. Related Works

### A. MRA-Based Pansharpening Methods

The traditional MRA-based methods perform pansharpening based on multiscale decomposition. The general form [24] can be expressed as follows:

$$\widehat{M}_k = \widetilde{M}_k + g_k(P - P_L) \tag{1}$$

where the subscript $k$ indicates the $k$th spectral band, $P_L$ is the low-frequency component of PAN image, and $g$ is the injection gain. $\widetilde{M}$ and $\widehat{M}$ denote the upsampled MS image and the fused HRMS image, respectively. Differences between MRA-based methods lie in the decomposition of $P_L$, and the estimation of $g_k$. MRA-based methods effectively preserve the entirety of the MS image content, ensuring spectral preservation [25].

Common solutions of $P_L$ include generalized Laplacian pyramid (GLP) [26], [27], wavelet transform [28], [29], contourlet transform [30], and curvelet transform [31]. However, given the discrete wavelet transform (DWT) is not shift invariable, which can lead to spatial distortions, the shift-invariable wavelet such as the "à trous" wavelet transform [32], [33] has been adopted to alleviate these issues. Various approaches have also been proposed to estimate $g_k$, including calculating ratios

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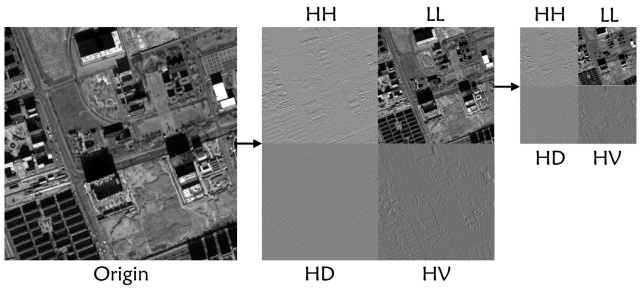



Fig. 3. Two-level 2-D DWT of a sample of PAN image from the SuperView-1 satellite.

between $\widetilde{M_k}$ and $P_L$, or applying multiple linear regression to them. Nonetheless, the estimation of $g_k$ is contingent upon the relationships between bands and images, which are challenging to establish artificially. DL networks, however, excel at discerning and leveraging these complex relationships, thereby mitigating the spatial distortion issues that are commonly associated with traditional MRA-based methods.

### B. DL-Based Multiscale Pansharpening Methods

In DL models, the receptive fields are expanded by the downsampling operations. The multiscale structure which combines deep and shallow features can simultaneously capture semantic representation and spatial details [34], [35], making it a widely used neural network architecture that has been extended to the field of pansharpening. Considering the inherent differences between PAN and MS images caused by their different sensors, researchers have proposed methods that extract PAN and MS features separately and fuse them at multiple levels. BDPN [36] extract PAN and MS features via a bidirectional pyramid network with original MS and PAN as inputs, and adds multiscale PAN features to corresponding MS feature maps; DFPN [37] employs four subnetworks to extract PAN and MS features at $2\times$ and $4\times$ scales, and refines spatial details through high-pass modulation modules; U2Net [38] extract spatial and spectral features by dual U-shape networks and equips spatial–spectral block for hierarchical fusion.

The methods mentioned above all perform pansharpening in the spatial domain, which typically focuses more on local features. In this work, motivated by the essence of pansharpening in the frequency domain, we integrate the concept of MRA-based methods with DL and utilize wavelet transform to progressively reconstruct HRMS images.

### C. Dual-Domain Analysis in DL-Based Pansharpening Methods

The combination of spatial and frequency domains can facilitate a comprehensive image analysis. Given the global nature of the frequency domain, recent works have applied frequency domain analysis to pansharpening. Most of them convert images to the frequency domain through the Fourier transform, then separately process and fuse the phase and amplitude (or real and imaginary parts) of PAN and MS images by convolution layers. In [39], researchers concatenated Fourier spatial and spectral units for global modeling of the difference between PAN and MS images. In [40], the Fourier transform is applied to the high-frequency image after Gaussian filtering. Zhou et al. [41] utilized a 2-D mask based on adaptive learning to separate high- and low-frequency components from the spectrum, thereby achieving effective reconstruction of high-frequency components. Unlike these approaches, our method employs DWT, which exhibits good locality in both spatial and frequency domains, to perform dual-domain operations, allowing for multiscale and precise frequency content analysis.

## III. Method

### A. Overview

The architecture of BiDFNet is shown in Fig. 2 which equipped D3F as the basic block. The network follows the bidirectional progressive fusion scheme, which adopts WAF modules for the progressive reconstruction of HRMS images. BiDFNet preserves the integrity of MS features and only performs wavelet transformation on PAN features for high-frequency extraction. This aligns with the core principle of the MRA-based methods.

### B. Discrete Wavelet Transform

According to the scaling space theory [42], the abundance of high-frequency information present in feature maps varies across different scales. For the task of pansharpening, extracting high-frequency components of PAN images at multiple scales facilitates a gradual refinement from a holistic to a localized perspective in high-frequency reconstruction, thereby reducing spatial and spectral distortions.

DWT, as a typical tool for MRA of images, utilizes scale functions to obtain a series of approximation components at different scales, while also using basic functions to extract high-frequency components. Take the 2-D Haar wavelet as an example. Given an image $I(x, y) \in \mathbb{R}^{H\times W\times C}$, the 2-D Haar wavelet decomposes it into an approximation component $LL \in \mathbb{R}^{(H/2)\times(W/2)\times C}$ and three high-frequency components $HH \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, $HV \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, and $HD \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, representing horizontal, vertical, and diagonal details, respectively. The inverse DWT (IDWT) combines the approximation and the high-frequency components from the current scale to reconstruct the approximation at the next higher level. In theory, DWT and IDWT are lossless, and capable of completely reconstructing the original image. Fig. 3 shows the decomposed components obtained after two wavelet transformations of a PAN image. Given the similar energy distribution between high-frequency in the HRMS and PAN images, the HRMS image can be obtained through wavelet reconstruction by combining these components extracted from the PAN image with the MS image, under suitable injection gains. Based on the above analysis, we combine wavelet transform with neural networks to achieve progressive pansharpening.

### C. Bidirectional Progressive Fusion Scheme

The bidirectional progressive fusion scheme consists of two subnets in opposite directions, which take MS and PAN images as inputs, respectively. The reverse subnet extracts PAN

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features and applies wavelet transform to decompose them into low-frequency and high-frequency components at multiple scales. Haar wavelet is employed for decomposing, replacing the traditional pooling layers. The decomposed low-frequency and high-frequency components at each scale are denoted as $P_n^L$ and $P_n^H$ respectively, where $n = 1, 2, \ldots, N$ indicates the number of DWT levels in the reverse subnet. $P_n^L$ undergoes further decomposition after processed by a D3F, yielding features with even lower resolution. Concurrently, the decomposed components are fed into the forward subnet. After $N$ iterations of wavelet decomposition, PAN features across $N+1$ different scales can be acquired. In the forward subnet, the MS features $M_n$ are sequentially fused with $P_n$ via WAF to reconstruct HRMS features with twice the resolution of $M_n$. Considering that the resolution of PAN images is commonly four times greater than that of MS images, we set $N = 2$ in our work. The multiscale MS features generated in the reconstruction process are integrated through a convolution layer, yielding the HRMS image with high spatial and spectral fidelity.

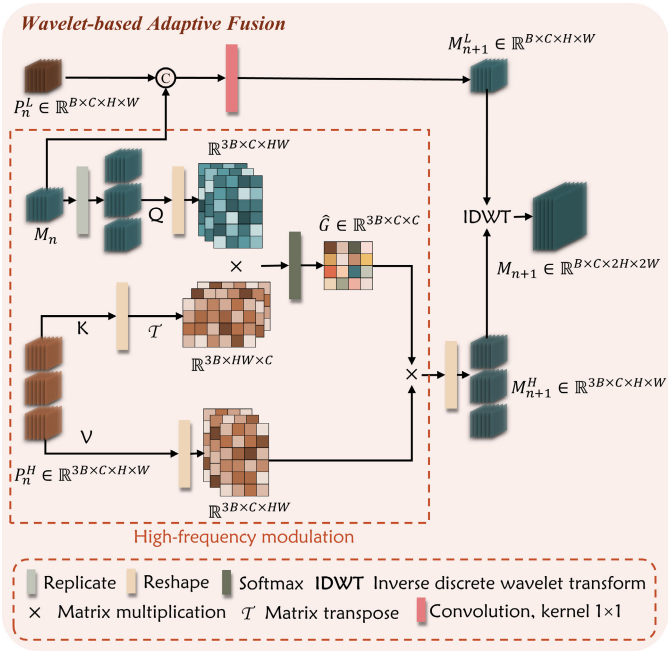



Fig. 4. Structure of WAF.

### D. Wavelet-Based Adaptive Fusion

From the frequency domain perspective, filling the decomposed high-frequency components of PAN features into the MS features is key to pansharpening. In this study, we adopted IDWT to accomplish this process. Given the distinct sensor sources for PAN and MS images, the high-frequency components of PAN lack color information specific to individual bands and need to be modulated by injection gains before being combined with MS features. However, traditional manually set gains cannot effectively parse the implicit relationships between spectral bands and spatial details, leading to degradation of the fused image. Therefore, we propose the WAF module to optimize the reconstruction of high-frequency features in HRMS by exploring the spatial–spectral dependence between PAN and MS features.

The structure of the WAF module is depicted in Fig. 4, it takes the decomposed components $P_n^H$ and $P_n^L$ of PAN features, and the MS features $M_n$ as inputs, which are at the same scale level. The band-specific injection gains used for modulating $P_n^H$ are learned from the correlation between $P_n^H$ and $M_n$ established by cross-attention.

With the replicated $M_n$ as Query ($Q$) features, and $P_n^H$ as Key ($K$) and Value ($V$) features, we flatten them along the spatial dimension, i.e., $\mathbb{R}^{C\times H\times W} \rightarrow \mathbb{R}^{C\times HW}$. The cross attention, taking $Q$, $K$, and $V$ as inputs, is as follows:

$$\begin{aligned} G &= \frac{QK^{\mathrm{T}}}{\sqrt{d_K}} \\ \hat{G} &= \mathrm{Softmax}(G) \\ H &= \hat{G}V \end{aligned} \tag{2}$$

where $G \in \mathbb{R}^{C\times C}$ indicates the relevance matrix between $C$ channels in $Q$ and $K$, which is divided by a scale factor $\sqrt{d_K}$ to enhance training stability. $d_k$ denotes the dimension of $K$. Given that the channel dimension in MS features represents multiband spectral information, $G$ contains the spatial–spectral relevance between the PAN and MS images along the spectral dimension, thereby learning customized modulation coefficients for each spectral band. By normalizing $G$ via softmax operation, the band-specific injection gain matrix $\hat{G}$ can be obtained. Notably, the band-specific injection gains are dynamic and adjusted based on the input images. This adjustment allows for the generalization of various scenarios and data sources. By modulating $\hat{G}$ to $V$, the cross attention aligns the spatial and spectral features by enhancing relevant features and suppressing redundant ones, yielding accurate high-frequency components utilized to reconstruct HRMS features.

The reconstructed high-resolution MS feature $M_{n+1}$ can be gotten as follows:

$$M_{n+1} = \mathrm{IDWT}\left(\mathcal{A}\left(P_n^H, \mathcal{R}(M_n)\right), \mathcal{C}\left(\left[P_n^L, M_n\right]\right)\right) \tag{3}$$

where $\mathcal{A}(\cdot)$ and $\mathcal{C}(\cdot)$ indicate cross attention and convolution layer respectively. Before computing the cross attention, the $M_n$ are first replicated along the batch dimension to match the shape of $P_n^H$, thus $\mathcal{R}(\cdot)$ denotes replication. The IDWT integrates high-frequency components from multiple directions with low-frequency features. By providing them with dynamic injection gains, WAF can precisely control detail information in each direction during reconstruction, ensuring that each scale level receives more accurate processing and reconstruction, thereby achieving clearer fusion results.

### E. Dual-Domain Dynamic Filtering

In practical application, interscene and intrascene variance [43] are important factors that impact the quality of pansharpening outputs. Although the inductive biases inherent in CNNs enhance the generalization capabilities of DL models, pansharpening in complex scenarios requires data-specific processes to increase the method's adaptability. Dynamically adjusting the convolutional weights offers a feasible solution by calculating the convolution parameters based on the

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input image. It allows for parameter adjustments during the inference stage, adapting more effectively to varying scenarios. Furthermore, the inclusion of global representations enables the model to comprehend the overall structure and inter-pixel relationships, thereby enhancing both spatial and spectral fidelity. The convolution theorem states that convolution in the spatial domain is equivalent to Hadamard production in the frequency domain. Consequently, global convolution in the spatial domain can be achieved by Hadamard production between a global filter and features in the frequency domain [44], [45]. In this work, we introduce the D3F to establish global and local representation by data-specific dynamic filtering in both frequency and spatial domains. The D3F effectively addresses the trade-off between model complexity and computation cost, thereby enhancing the representation ability of the model. The details are shown in Fig. 5.

For the input feature $F_{\text{in}}$, the D3F establishes short- and long-distance relationships through spatial and frequency branches, respectively. In the spatial branch, dynamic spatial filtering (DSF) is implemented via a convolution layer with learnable dynamic spatial filters denoted as $K_s^{C_{\text{out}}\times C_{\text{in}}\times k\times k}$ is proposed, which can dynamically establish local relationships. $k$ denotes the kernel size, $C_{\text{out}}$ and $C_{\text{in}}$ represent channels of the input and output feature maps respectively.

Fig. 5(c) illustrates the generation of $K_s$. $F_{\text{in}}$ is first squeezed along the channel dimension by a $1 \times 1$ convolution layer and subsequently limited to the range [0,1] by a sigmoid function. A spatial weight matrix $W_s \in \mathbb{R}^{H\times W\times 1}$ with inter-pixel relationships embedded in it is thereby obtained. Then $K_s$ is generated by modulating the weight of conventional convolution kernel $K_0$ with $W_s$, shown as follows:

$$K_s = W_s \odot W_k \tag{4}$$

where $W_k$ denotes the weight of conventional convolution kernel $K_0$, and $\odot$ indicates the Hadamard product of matrix.

In the frequency branch, we implement dynamic frequency filtering (DFF) for global convolution, which is designed based on the Fourier neural operators (FNOs) [46]. As is shown in Fig. 5(a), fast Fourier transform (FFT) is adopted to map the input feature $F_{\text{in}}$ into frequency domain along the spatial dimension

$$F_{\text{fre}}(u,v) = \frac{1}{HW}\sum_{x=0}^{W-1}\sum_{y=0}^{H-1} F_{\text{in}}(x,y)e^{-j2\pi(ux/W+vy/H)} \tag{5}$$

where $F_{\text{fre}}$ is the frequency-domain feature of $F_{\text{in}}$ and is a complex tensor. According to the convolution theorem, $F_{\text{fre}}$ is then modulated by a learnable global filter as follows:

$$\widetilde{F}_{\text{fre}} = K_f \odot F_{\text{fre}} \tag{6}$$

$$K_f = \mathcal{MLP}(F_{\text{fre}}) \tag{7}$$

where $K_f$ denotes the learnable dynamic global filter, which can be learned from $F_{\text{fre}}$. $\mathcal{MLP}(\cdot)$ represents a two-layer multilayer perceptron (MLP) in the frequency domain, as in [47] and [48]. It should be noted that the MLP shares weights and bias in spatial dimension and captures dependence between channels, so it is resolution-independent and lightweight. Then, the modulated features are mapped back to the spatial domain by iFFT and added with the original features $F_{\text{in}}$. Since $K_f$ is in the same shape as $F_{\text{in}}$, it equals a global filter in the spatial domain.

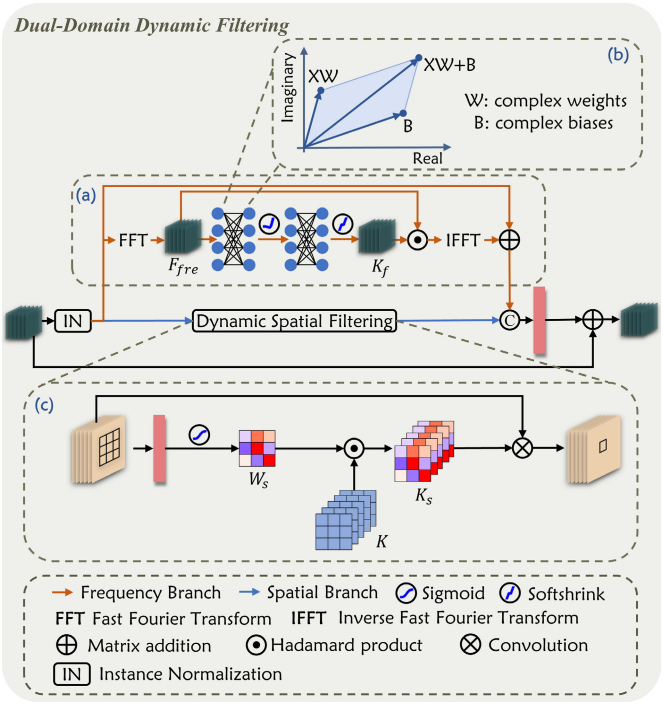



Fig. 5. Structure of D3F. (a) Workflow of DFF. (b) Diagram of MLP in the frequency domain involves complex multiplication. (c) Workflow of DSF.

### F. Loss Function

We trained the proposed method by minimizing the $\mathcal{L}_1$ loss function between the fusion results and the ground-truth images. The formula is as follows:

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N}\left\|\widehat{M}_i - \text{GT}_i\right\|_1 \tag{8}$$

where $N$ is the number of samples, and $\|\cdot\|_1$ is the L1 norm. $\widehat{M}$ and GT denote the output and ground-truth images, respectively.

## IV. Datasets and Experimental Details

### A. Datasets

We evaluate images acquired by five satellite sensors, including QuickBird, GaoFen-2, WorldView-III, WorldView-II, and SuperView-1. The parameters of these sensors are listed in Table I. The first three datasets are obtained from publicly available pansharpening datasets PanCollection [49]. The WorldView-II datasets is derived from the PairMax dataset [50], where we cropped the images in the FR folders into patches. Specific details of these datasets are provided in Table I.

The satellite images provided by Superview-1 were gathered between January and November 2019, covering an area of 896 km$^2$ in Beijing City and Zhangjiakou City. We follow the data preparation procedure outlined in [50] to generate the SuperView-1 datasets.

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TABLE I
DETAILS OF SATELLITE SENSORS AND DATASETS

| Satellite Sensors | | SuperView-1 | QuickBird | GaoFen-2 | WorldView-III | WorldView-II |
|---|---|---|---|---|---|---|
| Ground Sample Distance | PAN | 0.5m | 0.61m | 0.8m | 0.31m | 0.46m |
| | MS | 2m | 2.44m | 3.2m | 1.24m | 1.85m |
| Number of Multi-Spectral Bands | | 4 | 4 | 4 | 8 | 8 |
| Dynamic Range | | 11bit | 11bit | 10bit | 11bit | 11bit |
| Training Set | Sample Number | 18126 | 17139 | 19809 | 9714 | / |
| | Image Size | PAN 128×128×1 | PAN 64×64×1 | PAN 64×64×1 | PAN 64×64×1 | / |
| | | MS 32×32×4 | MS 16×16×4 | MS 16×16×4 | MS 16×16×8 | / |
| Validation Set | Sample Number | 2015 | 1905 | 2201 | 1080 | / |
| | Image Size | PAN 128×128×1 | PAN 64×64×1 | PAN 64×64×1 | PAN 64×64×1 | / |
| | | MS 32×32×4 | MS 16×16×4 | MS 16×16×4 | MS 16×16×8 | / |
| Reduced Resolution Testing Set | Sample Number | 115 | 20 | 20 | 20 | / |
| | Image Size | PAN 512×512×1 | PAN 256×256×1 | PAN 256×256×1 | PAN 256×256×1 | / |
| | | MS 128×128×4 | MS 64×64×4 | MS 64×64×4 | MS 64×64×8 | / |
| Full Resolution Testing Set | Sample Number | 27 | 20 | 20 | 20 | 15 |
| | Image Size | PAN 512×512×1 | PAN 512×512×1 | PAN 512×512×1 | PAN 512×512×1 | PAN 512×512×1 |
| | | MS 128×128×4 | MS 128×128×4 | MS 128×128×4 | MS 128×128×8 | MS 128×128×8 |
| Training Batch Size | | 32 | 32 | 32 | 32 | / |

Specifically, for reduced resolution experiments, to simulate the training samples under Wald's protocol [51], original MS and PAN images were first filtered by a Gaussian kernel that was generated according to the sensor's MTF and downsampled by a factor 4. The MS images before downsampling were treated as references. Then, the images were cropped and divided into a training set, a validation set, and a testing set, respectively. It is worth noting that the ms images in QuickBird, GaoFen-2, and SuperView-1 datasets consist of four bands(red, green, blue, and near infrared-1), while four new bands are acquired by WorldView-III and WorldView-II sensors(coastal, yellow, red edge, and near-infrared 2).

### B. Benchmark

We conduct quantitative and qualitative comparisons of BiDFNet with traditional and SOTA DL-based methods on the test sets. Five traditional methods are selected, including two CS-based methods (i.e., Brovey [52], adaptive Gram–Schmidt (GSA) [53]), two MRA-based methods (i.e., Wavelet [54] and smoothing filter-based intensity modulation (SFIM) [55]), and VO-based methods (i.e., CNMF [56]). Seven DL-based methods (i.e., PanNet, BDPN, FusionNet [57], INNTrans [58], LPPN [59], FAFNet [60], and DISPNet [61]) were selected. The DL-based methods are implemented following the default configuration pointed out in the related papers or source codes.

### C. Implement Details

BiDFNet is implemented in the PyTorch framework. During the training process, the learning rate is set as $1 \times 10^{-3}$ with the Adam optimizer. For a fair comparison, all the DL-based methods are trained on a desktop computer equipped with a single NVIDIA RTX 3090 GPU.

### D. Quality Assessment Indexes

To quantitatively evaluate the effectiveness of BiDFNet, several widely used image quality assessment indexes are selected. In the reduced resolution experiments, the structure similarity index measure (SSIM), the spectral angle mapper (SAM), the erreur relative globale adimensionnelle de synthèse (ERGAS), the spatial correlation coefficient (SCC), and $Q2^n$ index [62] are adopted, provided that the references are available. In the full-resolution experiments, quality assessment without reference is used. According to the analysis results in [63], the spectral distortion index ($D_\lambda^R$), the spatial distortion index ($D_s^R$), and the regression-based quality with no reference (RQNR) [64] index are exploited in this article. The ideal values for SSIM, SCC, and RQNR are 1, while for SAM, ERGAS, $D_\lambda^R$, and $D_s^R$ are 0.

## V. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Reduced Resolution Experiment

We first evaluated BiDFNet on the simulated reduced-resolution datasets for different sensors. The results are based on the test sets.

*1) Quantitative Analysis:* Table II exhibits the quantitative results of all the comparison methods on the test sets of GaoFen-2, SuperView-1, WorldView-III, and QuickBird datasets. The optimal values are indicated in bold, and suboptimal values are underscored. It can be seen that BiDFNet achieves the best performance in all indexes at datasets with different spatial and spectral resolutions. Especially the WorldView-III datasets, which is a more challenging case for pansharpening since the MS bands cover a wider range of wavelengths than the PAN band. And DISPNet shows suboptimal results.

*2) Qualitative Analysis:* To intuitively analyze the effectiveness of BiDFNet, we visualize the pansharpening samples of all the compared methods on QuickBird, GaoFen-2, SuperView-1, and WorldView-III datasets in Figs. 6–9, respectively. We also present the difference maps between pansharpening results and the reference image, highlighting the reconstruction performance on high-frequency details.

The sample of the QuickBird satellite contains sea surface, coastline, and vegetation, as depicted in Fig. 6, and the traditional method has obvious ambiguity in the boundary delineation. Among them, Brovey and GSA, as CS-based methods, retain the spatial structure during pansharpening,

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TABLE II
QUANTITATIVE EVALUATION RESULTS ON REDUCED RESOLUTION SAMPLES IN GAOFEN-2 (GF2), SUPERVIEW-1 (SV), QUICKBIRD (QB), AND WORLDVIEW-III (WV3) DATASETS

| Dataset | Method | Brovey | GSA | Wavelet | SFIM | CNMF | PanNet | BDPN | FusionNet | INNTrans | LPPN | FAFNet | DISPNet | Ours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| GF2 | SSIM | 0.9467 | 0.9560 | 0.9290 | 0.9569 | 0.9593 | 0.9315 | 0.9744 | 0.9760 | 0.9610 | 0.9883 | 0.9847 | 0.9895 | **0.9905** |
| | SAM | 0.0323 | 0.0301 | 0.0336 | 0.0290 | 0.0355 | 0.0315 | 0.0213 | 0.0256 | 0.0352 | 0.0164 | 0.0171 | 0.0145 | **0.0138** |
| | ERGAS | 2.3021 | 1.7231 | 2.1242 | 1.6552 | 1.7675 | 2.1879 | 1.2467 | 1.3558 | 2.0584 | 0.8279 | 0.9409 | 0.7518 | **0.7280** |
| | SCC | 0.6116 | 0.6120 | 0.4649 | 0.5934 | 0.6001 | 0.2668 | 0.6133 | 0.6519 | 0.6209 | 0.7184 | 0.6958 | 0.7535 | **0.7721** |
| | $Q2^n$ | 0.7833 | 0.8820 | 0.8279 | 0.8950 | 0.8843 | 0.8409 | 0.9454 | 0.9448 | 0.8757 | 0.9723 | 0.9719 | 0.9787 | **0.9808** |
| SV | SSIM | 0.8839 | 0.8795 | 0.8324 | 0.8917 | 0.8954 | 0.9320 | 0.9582 | 0.9492 | 0.9565 | 0.9835 | 0.9828 | 0.9856 | **0.9870** |
| | SAM | 0.0330 | 0.0355 | 0.0403 | 0.0316 | 0.0394 | 0.0303 | 0.0228 | 0.0297 | 0.0286 | 0.0176 | 0.0161 | 0.0156 | **0.0148** |
| | ERGAS | 2.9001 | 2.8821 | 3.0263 | 2.3333 | 2.7243 | 1.8033 | 1.3307 | 1.7110 | 1.7147 | 0.7985 | 0.8139 | 0.7361 | **0.6969** |
| | SCC | 0.1742 | 0.1734 | 0.1175 | 0.1516 | 0.1666 | 0.3643 | 0.5553 | 0.5485 | 0.6134 | 0.7192 | 0.7116 | 0.7769 | **0.7802** |
| | $Q2^n$ | 0.7456 | 0.8020 | 0.7312 | 0.8430 | 0.8173 | 0.8978 | 0.9452 | 0.8813 | 0.9235 | 0.9735 | 0.9758 | 0.9809 | **0.9825** |
| QB | SSIM | 0.8128 | 0.8351 | 0.8076 | 0.8276 | 0.8498 | 0.7317 | 0.9335 | 0.8809 | 0.9334 | 0.9223 | 0.9293 | 0.9401 | **0.9472** |
| | SAM | 0.1493 | 0.1459 | 0.1659 | 0.1420 | 0.1358 | 0.1441 | 0.1006 | 0.1092 | 0.0991 | 0.0956 | 0.0954 | 0.0949 | **0.0943** |
| | ERGAS | 9.5073 | 8.5096 | 9.0076 | 8.5894 | 8.0720 | 11.1620 | 5.0183 | 7.8739 | 5.2142 | 5.7705 | 5.5133 | 4.9467 | **4.7266** |
| | SCC | 0.5404 | 0.5373 | 0.4462 | 0.5426 | 0.5333 | 0.4071 | 0.6894 | 0.5098 | 0.7036 | 0.6200 | 0.6404 | 0.7138 | **0.7185** |
| | $Q2^n$ | 0.7249 | 0.7819 | 0.7686 | 0.7838 | 0.8019 | 0.6446 | 0.9047 | 0.8816 | 0.8972 | 0.8882 | 0.8960 | 0.9184 | **0.9259** |
| WV3 | SSIM | 0.8486 | 0.8635 | 0.8201 | 0.8670 | 0.8636 | 0.8670 | 0.9552 | 0.9605 | 0.9561 | 0.9672 | 0.9671 | 0.9719 | **0.9722** |
| | SAM | 0.1018 | 0.0972 | 0.1206 | 0.0932 | 0.0976 | 0.1070 | 0.0685 | 0.0648 | 0.0760 | 0.0597 | 0.0564 | 0.0522 | **0.0518** |
| | ERGAS | 5.6025 | 5.1785 | 5.8221 | 4.9804 | 5.0813 | 5.0265 | 2.9075 | 2.7313 | 2.9577 | 2.4814 | 2.4436 | 2.2276 | **2.2103** |
| | SCC | 0.5625 | 0.5630 | 0.4447 | 0.5648 | 0.5601 | 0.4762 | 0.6890 | 0.7042 | 0.6974 | 0.7070 | 0.7108 | 0.7303 | **0.7393** |
| | $Q2^n$ | 0.6954 | 0.7886 | 0.6852 | 0.8089 | 0.7697 | 0.7648 | 0.8901 | 0.8902 | 0.8660 | 0.8933 | 0.9081 | 0.9133 | **0.9173** |

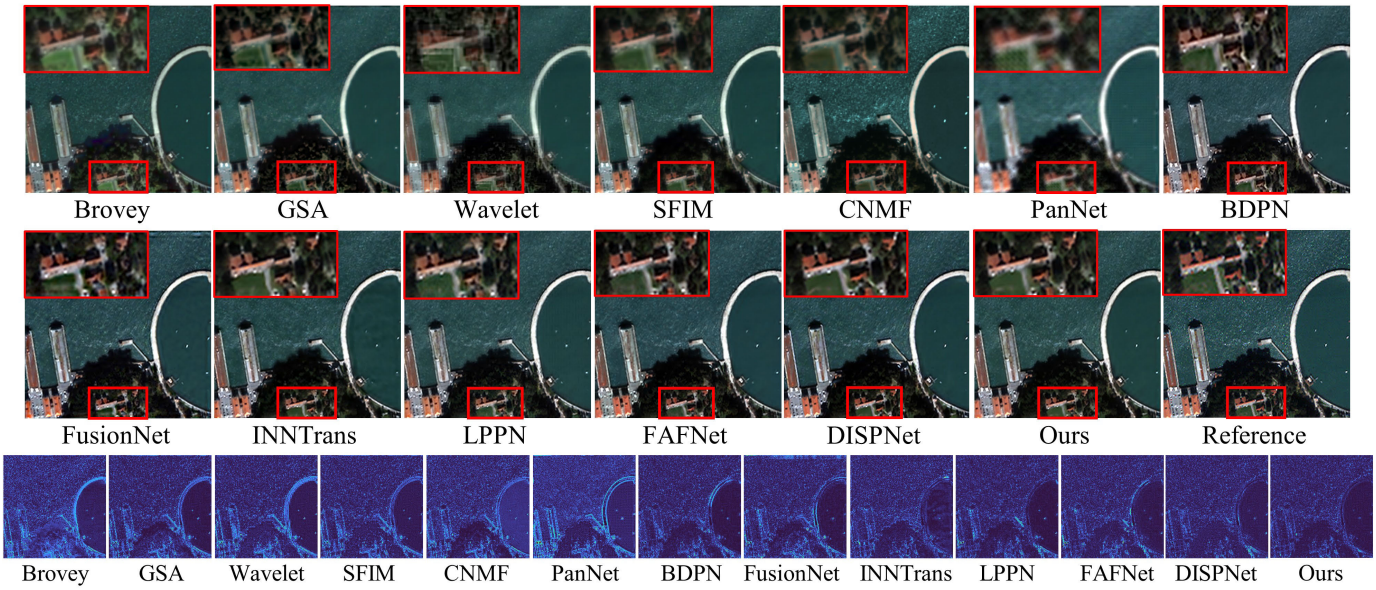



Fig. 6. Qualitative comparisons on the sample of the reduced resolution QuickBird datasets. The second and fourth lines are difference maps. The box selection area is enlarged for a clearer display.

while the wavelet yields noticeable artifacts at the edges. And obvious spectral distortion occurs in the result of CNMF. In terms of the DL-based methods, PanNet results in spatial blurring on this sample, which may be associated with the inaccurate separation of high frequencies by the high-pass filter. The straight path in front of the houses is crooked by INNTrans, LPPN, and FAFNet, and the traditional methods all reconstruct it to the wrong color. What's more, the residual maps show noticeable misalignment in the lower right corner of the result of DISPNet.

Similar results are also depicted in the samples of GaoFen-2 and SuperView-1 datasets. As shown in Fig. 7, BDPN, which adopts a similar multiscale bidirectional network structure as our method, maintains the high fidelity of the spatial structure. While it doesn't explicitly establish spatial–spectral dependencies and causes spectral distortion on the blue roof. In contrast, INNTrans successfully preserved the spectral information but distorted the boundaries. LPPN, FAFNet, DISPNet, and BiDFNet yielded high spatial and spectral fidelity, and the residual maps show that the high-frequency component reconstructed by BiDFNet is closer to the real HRMS images. Furthermore, benefiting from the precise frequency analysis of wavelet transform, the dense mesh textures in Fig. 8 are reconstructed by BiDFNet without deformation, while other methods blur the edges.

Fig. 9 illustrates the sample from the WorldView-III datasets, which contains many small-scale features, such as densely distributed buildings (red box) and moving traffic flows (green box). These features pose challenges due to potential misalignment between spatial and spectral

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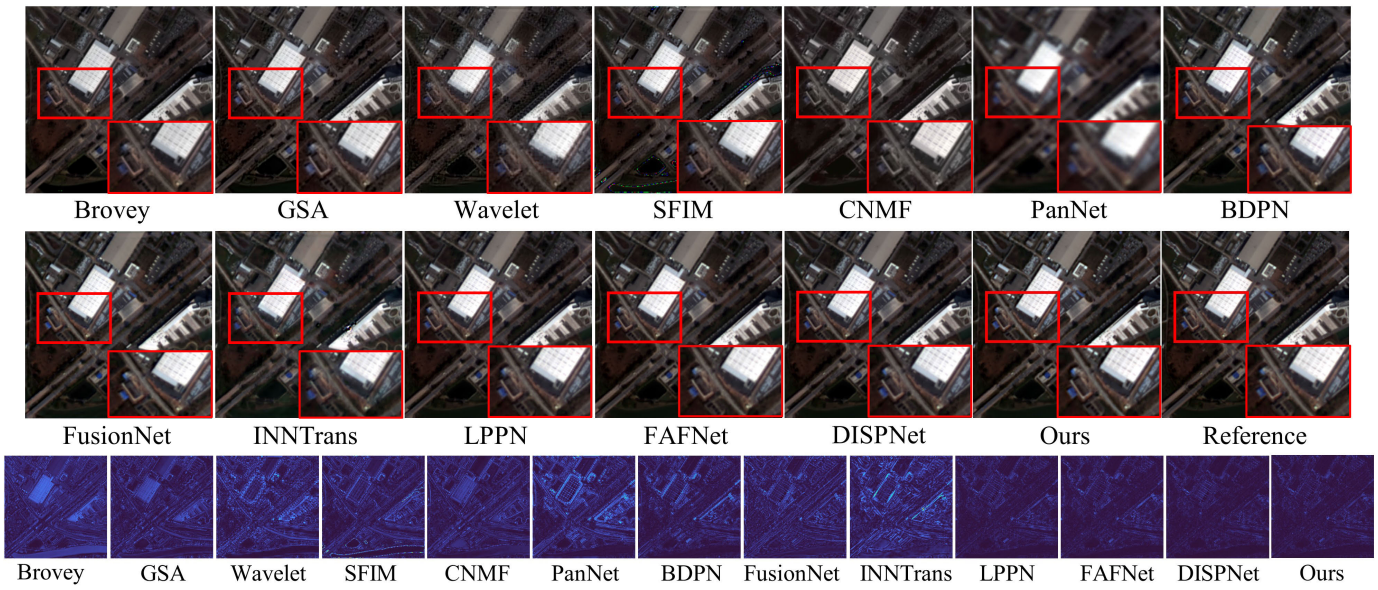



Fig. 7. Qualitative comparisons on the sample of the reduced resolution GaoFen-2 datasets. The second and fourth lines are difference maps. The box selection area is enlarged for a clearer display.

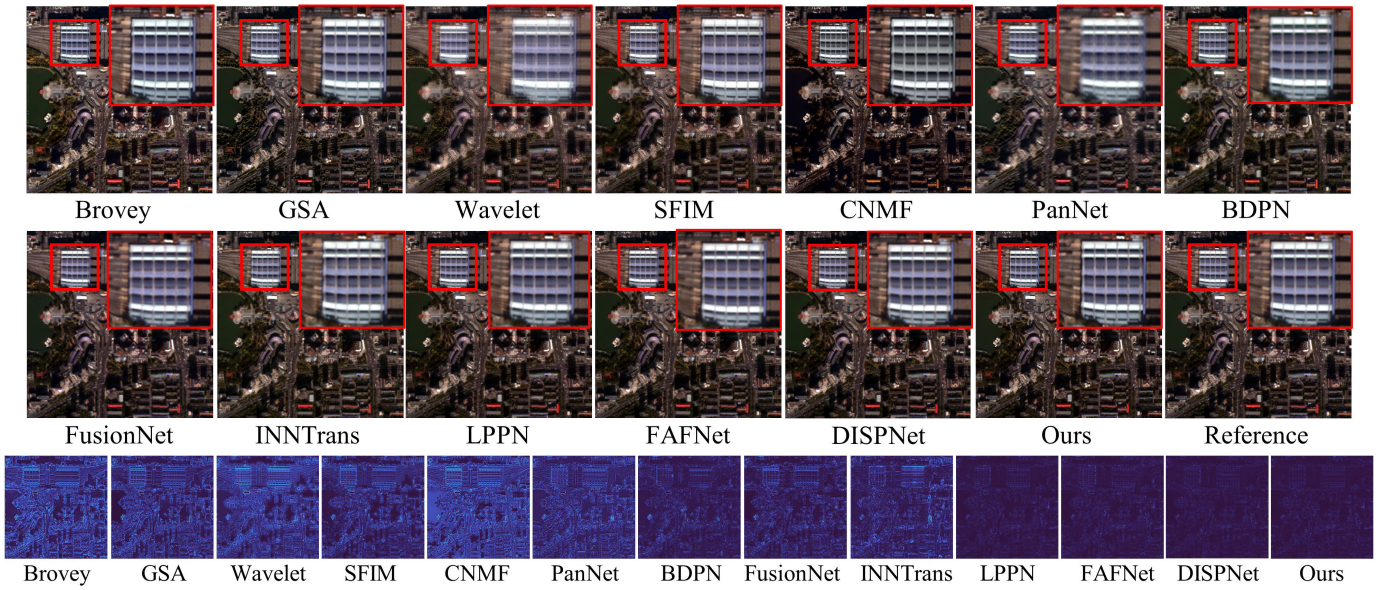



Fig. 8. Qualitative comparisons on the sample of the reduced resolution SuperView-1 datasets. The second and fourth lines are difference maps. The box selection area is enlarged for a clearer display.

components, which can significantly distort small targets in the reconstructed image. By applying proper injection gains to high-frequency components in multiple directions, BiDFNet preserves each vehicle's spatial and spectral independence. In contrast, the results from FAFNet, LPPN, and DISPNet show adjacent vehicles as a single entity. As for the roofs, BiDFNet maintains high color fidelity for blue roofs by establishing the spatial–spectral correlation. However, PanNet, BDPN, FusionNet, and INNTrans experience some color loss during the HRMS reconstruction.

### *B. Full-Resolution Experiment*

Considering that the spatial gap between the simulated reduced-resolution images and the origin images makes it challenging to test the pretrained models on full-resolution images, we also conducted experiments on the full-resolution datasets to evaluate the generalization and robustness of the comparison methods. Specifically, we select the QuickBird and WorldView-III datasets as the representations which have 4 and 8 bands of MS images, respectively. It should be noted that the result of INNTrans on the WorldView-III datasets appeared obvious color cast, for fair comparison, this method was not included in the full-resolution experiments.

The results on the sample from QuickBird datasets are illustrated in Fig. 10. In this sample, the characteristics of each class of traditional methods are more prominent. Brovey and GSA both achieve higher spatial fidelity; however, they also exhibit spectral distortion. The redundant high-frequency

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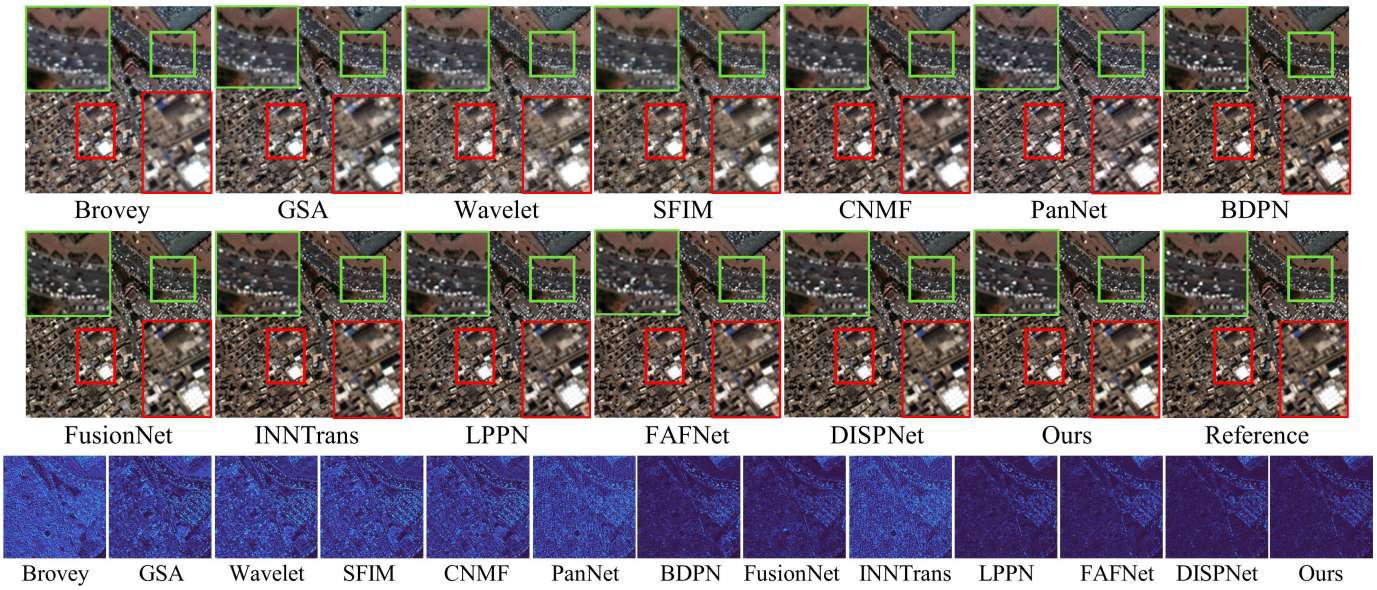



Fig. 9. Qualitative comparisons on the sample of the reduced resolution WorldView-III datasets. The second and fourth lines are difference maps. The box selection areas are enlarged for a clearer display.

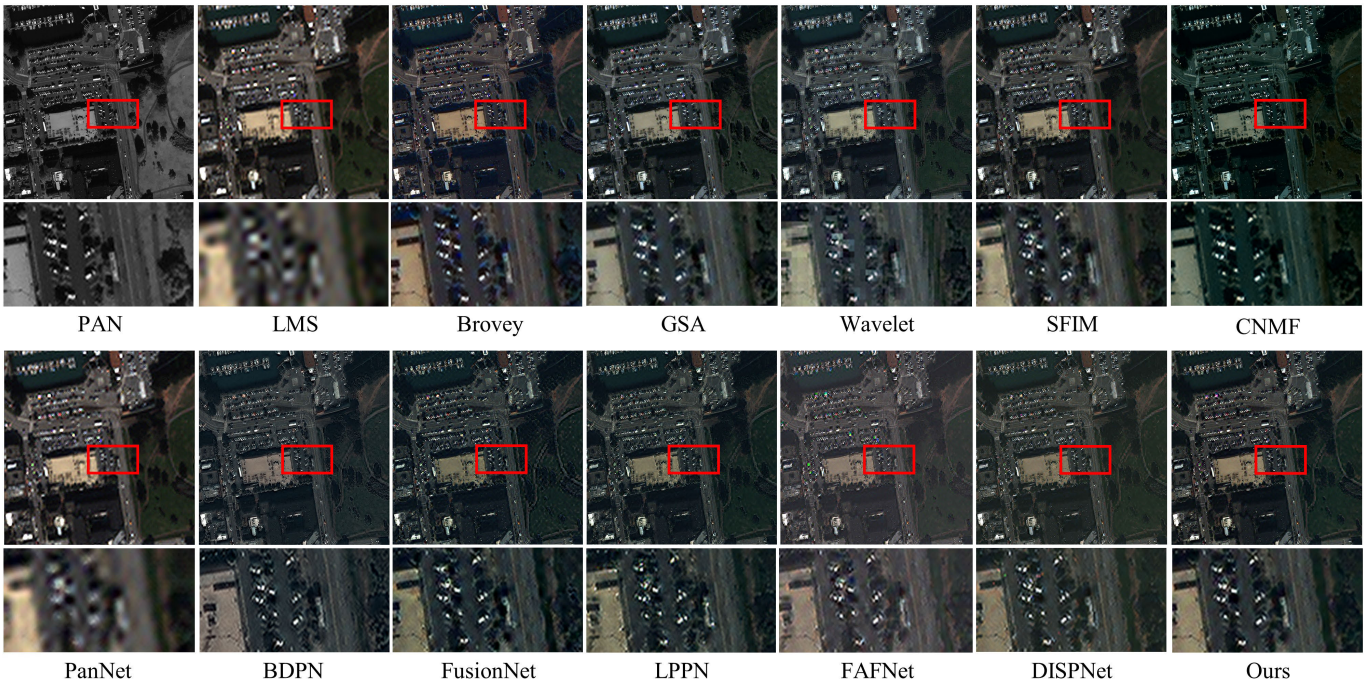



Fig. 10. Qualitative comparisons on the sample of the full-resolution QuickBird datasets. The box selection area is enlarged for a clearer display.

information in the wavelet method leads to noticeable spatial artifacts, and the SFIM method blurs edges due to its smoothing filter. Among DL-based methods, FusionNet. LPPN, and FAFNet cause artifacts around the vehicles with high-frequency aliasing, leading to a lack of clarity. BDPN and BiDFNet both take the original MS image as input and reconstruct high-frequency details, depending on the real information in PAN images. This strategy effectively avoids the pseudo-high-frequency interference caused by interpolation on MS images, facilitating high spatial fidelity. DISPNet achieves better spatial fidelity by adding internal supervision loss at various stages of the deep unfolding network. However, it suffers from excessive edge sharpening and spectral noise. Our method demonstrates superior performance in detail preservation, color balance, and artifact minimization.

Fig. 11 shows the pansharpening results on the sample from WorldView-III datasets. Overall, except for CNMF which causes a noticeable color cast, the other methods achieve good visual results. However, when the selected area region is magnified, it becomes evident that, except for BiDFNet, other DL methods fail to accurately align colors and boundaries when reconstructing the white roof, resulting in blurred edges. Similarly, significant artifacts and local noise appear in the results of LPPN, FAFNet, and DISPNet for small targets

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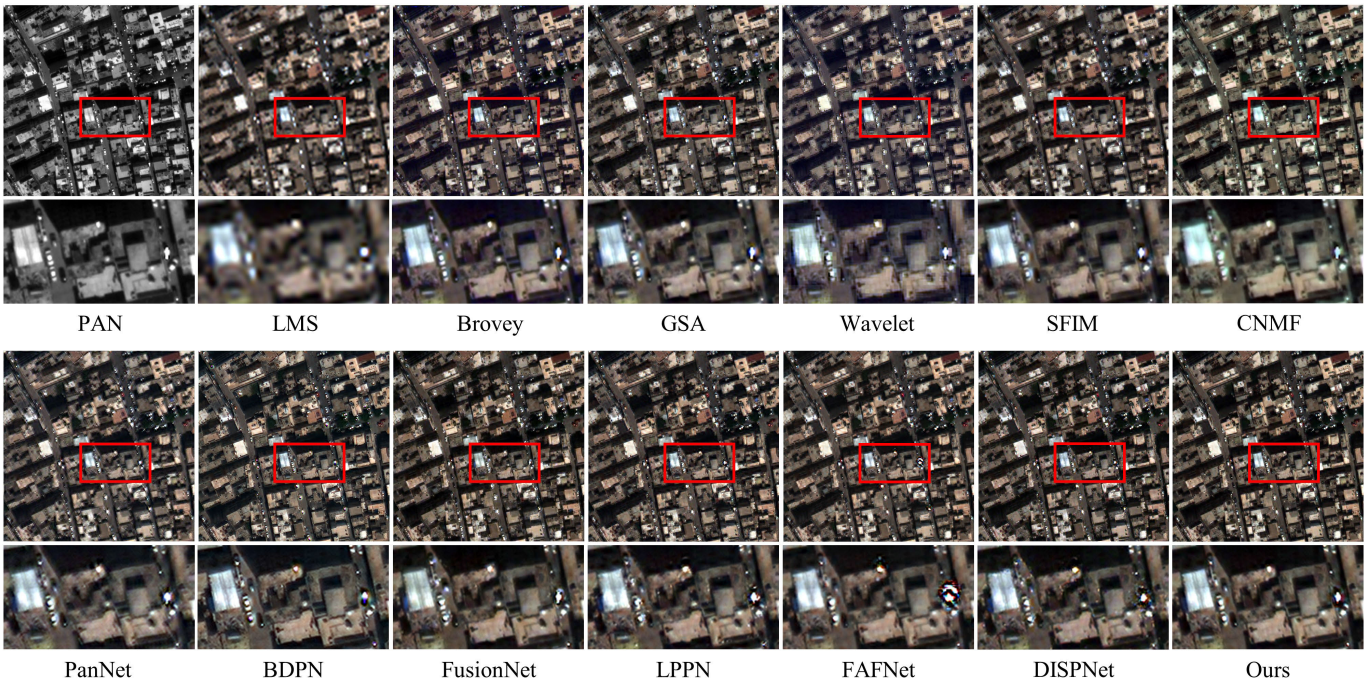



Fig. 11. Qualitative comparisons on the sample of the full-resolution WorldView-III datasets. The box selection area is enlarged for a clearer display.

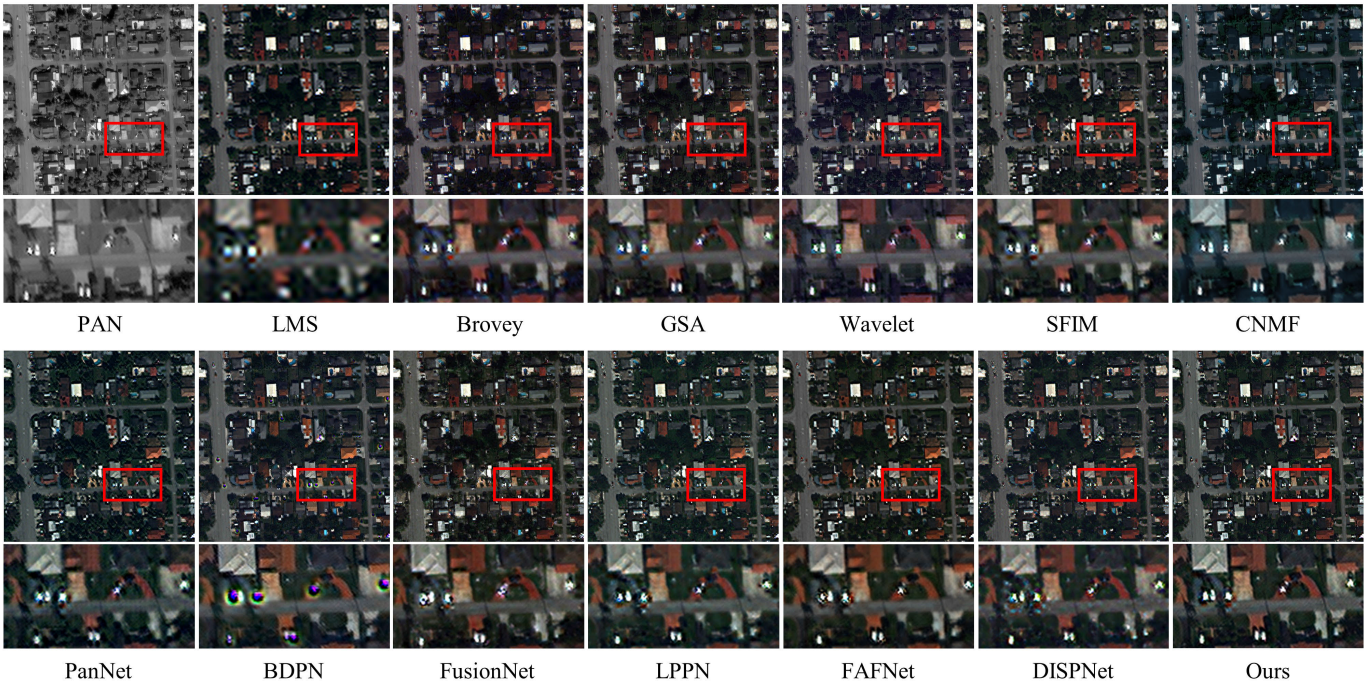



Fig. 12. Qualitative comparisons on the sample of the full-resolution WorldView-II datasets. The box selection area is enlarged for a clearer display.

such as vehicles. In contrast, BiDFNet, leveraging data-specific dynamic filtering, maintains its consistently excellent performance, successfully reconstructing vehicles with proper shapes and clear boundaries.

To further demonstrate the generalization of the proposed method, we conducted a more challenging experiment. We tested the models trained on WorldView-III datasets on WorldView-II datasets, which are acquired by different satellite sensors but have similar MS bands and spatial resolution. The visualized samples are presented in Fig. 12. BiDFNet achieves the best performance on both spectral and spatial fidelity.

The quantitative results on QuickBird, WorldView-III, and WorldView-II datasets are listed in Table III. Due to the rich spatial details in high-resolution imagery, local fidelity is crucial for subsequent tasks such as object recognition,

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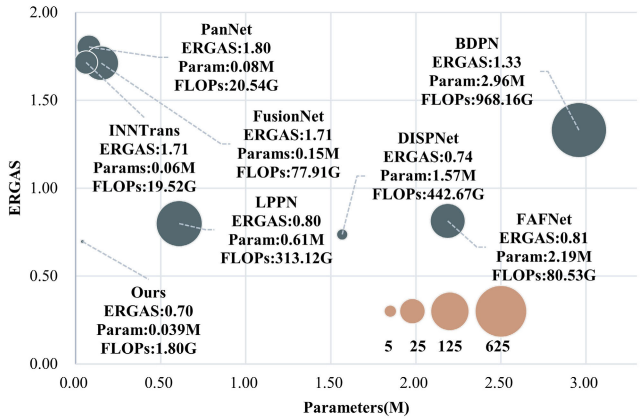



Fig. 13. Comparison of model complexity and ERGAS on the test set of the SuperView-1 dataset. The radius of circles represents the FLOPs.

detection, and extraction. Given this, RQNR is selected for evaluation. The RQNR protocol assesses the spectral fidelity by employing the consistency property of Wald's protocol as in [65] and [66], denoted as $D_\lambda^R$, which has been demonstrated effective for spectral quality evaluation on full-resolution data than other available strategies [63]. The spatial distortion index of RQNR denoted as $D_s^R$, excels in detecting local spatial distortion, especially those that do not significantly affect the statistical parameters but are still perceptually important. As reported in Table III, BiDFNet achieves the best overall balance between spectral and spatial preservation with the highest RQNR value across three datasets, indicating superior quality. While PanNet delivers high spectral quality, GSA and Brovey show the best spatial preservation with the lowest $D_s^R$ value.

### C. Comparison of Complexity

When selecting pansharpening methods in practice, it is essential to consider not only image quality but also the complexity of DL models. Less complex methods are more suitable for deployment on lightweight platforms, ensuring efficient image processing. To assess the complexity of BiDFNet, we analyze the floating point operations (FLOPs) and parameter numbers (Param) on the test set of the SuperView-1 dataset. The resolution of MS images in the dataset is $128 \times 128 \times 4$. The comparison of model complexity and ERGAS is shown in Fig. 13, where the radius of circles represents the FLOPs. Leveraging the proposed D3F as the basic block of the backbone, BiDFNet balances higher accuracy with reduced complexity.

### D. Ablation Study

To evaluate the effectiveness of the proposed modules, we conducted ablation experiments by removing or replacing specific modules. According to the results in Section V, the performance of the algorithm on the reduced resolution data and the full-resolution data can be very different, which is not conducive to practical application. Therefore, to truly reflect the effectiveness of each module, the ablation experiments were carried out on the full-resolution test set in WorldView-III datasets.

We conducted four ablation versions of our method, denoted as Ablation 1-4. The visualized samples of results are presented in Fig. 14 for a more clear and intuitive illustration, and the quantitative results are listed in Table IV.

*1) Dual-Domain Adaptive Filtering:* The D3F integrates global and local features, enabling the network to capture the overall structure of the image while focusing on local details. Due to the mapping of input features into spatial and frequency domain filters, the network possesses adaptive capabilities toward the input data. We conducted three experiments to validate the necessity of D3F. The first two experiments replaced the DSF and DFF with a conventional $3 \times 3$ convolution layer, respectively. Then, the third experiment replaced all D3F blocks with ResBlocks. These three variations are denoted as Ablation 1, 2, and 3, respectively. In Fig. 14, Ablation 3 exhibits apparent spatial deformation and contour artifacts of vehicles, which are substantially mitigated in Ablation 1 and 2. This suggests that the dynamic filtering has enhanced the network's generalization, preserving structural integrity at both global and local scales. Furthermore, Ablation 1 reveals well-defined boundaries, while Ablation 2 presents enhanced accuracy in spectral information. These observations indicate that the DSF successfully captured the interpixel relationships and refined local details. Simultaneously, the DFF improved spectral fidelity by establishing interband dependence.

*2) Wavelet-Based Adaptive Fusion:* The WAF is designed to adaptively inject high-frequency features, decomposed from PAN features, into MS features. The injection gains are deduced from the spatial–spectral correlation between the high-frequency features and MS features, established by the cross-attention mechanisms. Based on Ablation 3, we substituted a $1 \times 1$ convolution layer for the cross attention in WAF and represented it as Ablation 4. As depicted in Fig. 14, obvious color deviation appears in Ablation 4, which is associated with the improper injection of high-frequency features. The WAF optimal the injection by modulating high-frequency features with data-adaptive gains, thereby enhancing the model's generalization capabilities to novel datasets.

### E. The Number of DWT Levels

In this article, the wavelet transform is adopted to decompose and fuse the high-frequency components missing in the low-resolution MS image, which halves the spatial resolution of the decomposition components with each process. This means that the decomposition should stop when the PAN and upsampled MS image reach the same scale as the low-resolution MS image. At this point, the MS image's spectrum is restored to its initial state before upsampling. Further decomposition would damage the initial spectral information of the MS image, which is not conducive to pansharpening. This principle is consistent with traditional wavelet transform-based pansharpening studies. Therefore, the number of DWT levels $N$ is associated with the scale factor $S$ between PAN and MS images. In our study, the MS images are not upsampled, so the PAN images should be decomposed until the decomposition components match the MS scale, and the relationship between $N$ and $S$ is $S = 2N$.

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TABLE III
QUANTITATIVE EVALUATION RESULTS ON FULL-RESOLUTION SAMPLES IN QUICKBIRD DATASETS

| Method | QuickBird | | | WorldView-III | | | WorldView-II | | |
|---|---|---|---|---|---|---|---|---|---|
| | $D_\lambda^R$ | $D_s^R$ | RQNR | $D_\lambda^R$ | $D_s^R$ | RQNR | $D_\lambda^R$ | $D_s^R$ | RQNR |
| Ideal | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| Brovey | 0.4364 | 0.0177 | 0.3485 | 0.2209 | **0.0037** | 0.5398 | 0.2313 | 0.0039 | 0.5222 |
| GSA | 0.4100 | **0.0049** | 0.3831 | 0.2106 | 0.0176 | 0.5473 | 0.1840 | **0.0011** | 0.6067 |
| Wavelet | 0.3122 | 0.1611 | 0.4337 | 0.1917 | 0.0695 | 0.5403 | 0.1830 | 0.0336 | 0.5828 |
| SFIM | 0.2937 | 0.1542 | 0.4584 | 0.1853 | 0.0584 | 0.5594 | 0.1588 | 0.0479 | 0.6138 |
| CNMF | 0.4150 | 0.0059 | 0.3769 | 0.2231 | 0.0047 | 0.5354 | 0.2055 | 0.0047 | 0.5658 |
| PanNet | **0.0424** | 0.3252 | 0.6489 | **0.0158** | 0.0586 | 0.8886 | **0.0321** | 0.0954 | 0.8096 |
| BDPN | 0.1036 | 0.1624 | 0.6994 | 0.0393 | 0.0531 | 0.8439 | 0.0822 | 0.0874 | 0.7189 |
| FusionNet | 0.0886 | 0.2336 | 0.6655 | 0.0441 | 0.0628 | 0.8224 | 0.1285 | 0.0964 | 0.6252 |
| LPPN | 0.0633 | 0.2622 | 0.6757 | 0.0199 | 0.0672 | 0.8689 | 0.0377 | 0.1119 | 0.7791 |
| FAFNet | 0.0705 | 0.2410 | 0.6835 | 0.0259 | 0.0760 | 0.8455 | 0.0487 | 0.0723 | 0.8019 |
| DISPNet | 0.0632 | 0.2327 | 0.7001 | 0.0217 | 0.0621 | 0.8713 | 0.0409 | 0.1049 | 0.7807 |
| Ours | 0.1021 | 0.0703 | **0.7706** | 0.0251 | 0.0402 | **0.8904** | 0.0431 | 0.0601 | **0.8278** |

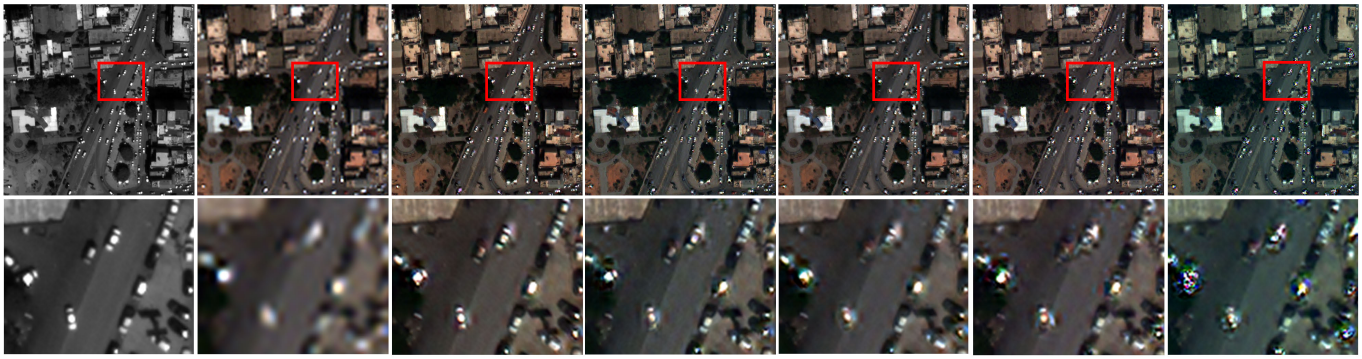



Fig. 14. Visualize results of ablation study on the sample of the full-resolution WorldView-III datasets. The box selection area is enlarged for a clearer display.

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF ABLATION STUDY AND DECOMPOSITION METHOD COMPARISON ON FULL-RESOLUTION SAMPLES IN WORLDVIEW-III DATASETS

| Method | Ideal | Ours(DWT) | Ablation 1 | Ablation 2 | Ablation 3 | Ablation 4 | WLS | LPL |
|---|---|---|---|---|---|---|---|---|
| $D_\lambda^R$ | 0 | 0.0251 | 0.0250 | 0.0319 | **0.0194** | 0.0221 | 0.0767 | 0.0322 |
| $D_s^R$ | 0 | **0.0402** | 0.0593 | 0.0664 | 0.0722 | 0.1194 | 0.0472 | 0.0495 |
| RQNR | 1 | **0.8904** | 0.8675 | 0.8442 | 0.8639 | 0.8013 | 0.7720 | 0.8636 |

### *F. Different Decomposition Methods*

In this work, the decomposition of PAN features is a key step. To verify the effectiveness of the wavelet transform adopted in our method, we compared it with two other decomposition methods: the LPP and the weighted least squares filter (WLS). The results, presented in Table IV, show that DWT achieves the best results, while WLS leads to a high spectral distortion index. This is likely because, despite preserving edges, WLS does not explicitly separate high-frequency details. It merges these details into a smoothed representation that prioritizes overall image coherence rather than specific frequency bands. Compared to WLS, LPP offers better separation of high-frequency details but operates primarily in the spatial domain. It focuses on the differences between multilevel images which are not explicitly tied to particular frequency ranges. In contrast, DWT captures specific frequency components at multiple levels, allowing for precise analysis and manipulation of the image's frequency content, which aligns well with the requirements of the pansharpening task as previously analyzed.

## VI. CONCLUSION

Injecting high-frequency details in the frequency domain allows for the selective enhancement or merging of specific frequency components, maintaining the global structure and preserving the spectral integrity of MS images. This ensures that the HRMS image generated by pansharpening benefits from rich details and accurate spectral information. To integrate frequency domain analysis with DL models, we proposed BiDFNet for high spatial and spectral fidelity pansharpening. With D3F blocks as basic blocks of the backbone, BiDFNet adopts wavelet transform to accomplish a bidirectional progressive fusion scheme. Multilevel high-frequency components are decomposed from PAN images by wavelet transform, and then progressively injected into MS features via WAF modules with data-specific gains. In both reduced-resolution

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and full-resolution experiments, the quantitative indexes and visualize results of BiDFNet are superior to other advanced methods. Ablation studies based on full-resolution data have demonstrated that WAF effectively enhanced spectral fidelity through learnable data-specific gains, while D3F maintained the integrity of spatial structures and improved the model's adaptability to complex scenes through dynamic filtering in both spatial and frequency domains. Furthermore, benefiting from mapping the input image into dynamic filters in D3F, BiDFNet achieves excellent generalization with low FLOPs and a small parameter count, providing the potential for subsequent deployment to different platforms. In the future, we will further study the full-resolution assessment metrics that better align with human visual perception and advance multitask learning research that integrates pansharpening with high-level tasks such as object detection and land cover mapping.

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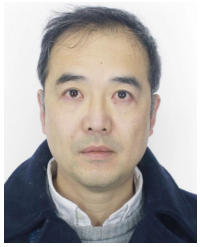
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